

2 LASER LIGHT TRANSFORMING VISION AND MACHINES

Usefulness of semiconductor laser technology is constantly increasing, especially quantum dot (QD) lasers that reliably operate with low noise at high temperatures. Vision is evolving through QD's laser light by precise light control to project images into the retina. It helps improve the low vision of people with disorders in the anterior parts of the eye, such as the lens or cornea.

QD lasers have superior environmental durability, which has led to the next revolution in silicon photonics. Superior high-temperature stability and optical feedback tolerance of QD lasers make them ideal light sources for silicon photonics. These have the potential to advance the world of communication by replacing electrical interconnects in circuits.

QD Laser Co. Inc. designs semiconductor laser-based solutions for a broad range of fields, including manufacturing, telecommunications, medical and consumer products. Their product range includes QD laser arrays for silicon optical circuits, broadband gain chips for ophthalmic testing equipment, picosecond short-pulse distributed feedback lasers for precision processing, current injection-type

green, yellow-green and orange lasers for biological sciences, and retinal imaging laser eyewear.

QD Laser offers 532nm, 561nm and 594nm wavelength semiconductor lasers. These innovative devices contribute to biomedical and sensor applications as well as miniaturisation and extended life of devices. When stained cells are irradiated with a laser, various cell information can be obtained from luminescence and light scattering. Ultra-high-resolution microscopes for bio-imaging combine excitation light for observation with torus-shaped STED light that suppresses surrounding blur. These provide observation at a much finer scale.

Ultra-short pulse distributed feedback lasers are used for laser machining, which allows non-thermal processing to remove target materials before heating. This laser allows micro-fabrication of any material and high-precision processing through programmed laser pulse intervals.

QD lasers are based on molecular beam epitaxy, which is a crystal growth technology, and support a diverse wavelength lineup of QD laser modules. QD lasers that use QDs in their active layer have excellent tem-

perature stability and energy efficiency. These can operate at 100°C with little performance deterioration and can be tailored to lase at 200°C. QD lasers can freely and precisely control emission wavelengths by making full use of grating formation technology embedded in semiconductor lasers.

"QD lasers serving as the light source for silicon optical circuits that process ultra-large volumes of information can expand the information-processing capabilities of systems with low power consumption. Our retinal imaging laser eyewear will likely be used in the future to assist people with vision problems, support ophthalmic and medical examinations," says Mitsuru Sugawara, president and chief executive officer, QD Laser Co. Inc.

The company's ultra-compact optical system projects light onto the retina to form a clear image almost irrespective of the focus position or the condition of the user's cornea or lens. This can also be used for augmented reality (AR) experiences. The system projects an image onto the retina that overlaps what is seen with the eye to create natural AR. Image at AR does not blur regardless of where the user is looking.

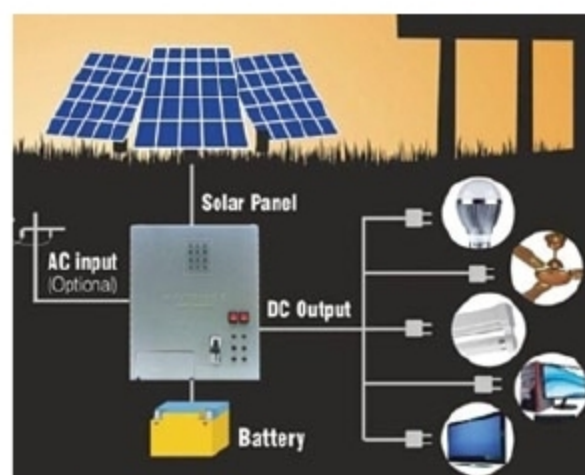
—Deepshikha Shukla

3 LIGHTING UP INDIA WITH DC POWER

Even after completing nearly two decades in the 21st century, a large population in developing and underdeveloped countries does not have access to constant power supply. Power lines are too costly for most of these places, while others have long power cuts. Although solar rooftops have grown to be the preferred option for providing decentralised power, high AC-DC conversion losses are common in photovoltaic systems besides their high cost.

Hyderabad-based startup Cygni Energy provides innovative DC-based solar solutions preventing the usual

need for multiple AC-DC conversions for both off-grid and grid-connected systems. Need for smaller panels and



(Credit: www.indiamart.com)

batteries leads to smaller panels and reduced battery costs. DC technology can be integrated with AC power supply, and this ensures uninterrupted supply even during load-shedding. For this, a DC line can be easily deployed inside a building with little change to the power supply system.

The company offers a broad range of variants. Their innovative integrated battery inverter-less system, Inverterless S, switches to the battery on its own in case of power outage. It is 500VA solar AC equivalent with a load capacity of 500W and five hours backup. Venkat Rajaraman,